

Geneva Ecola

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Education

Expected 2027	Ph.D. Stanford University , Electrical Engineering <i>Thesis:</i> Harnessing Ambient Radio Signals for Ultra-Low Power Connectivity <i>Advisor:</i> Zerina Kapetanovic, <i>Thesis Committee:</i> Howard Zebker, Joseph Kahn
2023	M.S. Stanford University , Electrical Engineering
2019	B.S. Harvey Mudd College , Engineering

Research Experience

2023 – Present	Stanford Smart Sensing Systems Lab , Doctoral Researcher Working on new ultra-low-power forms of communication including satellite backscatter connectivity and thermal noise backscatter communication.
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Publications

(*indicates co-first authorship)

Under Review	First-Principles Channel Model and Capacity-Achieving Transmission Scheme in Thermal Noise Backscatter Links <u>Geneva Ecola*</u> , Jida Zhang*, Joesph Kahn and Zerina Kapetanovic
2025	Wireless Johnson Noise Thermometry for Passive Temperature Sensing Jasmin Falconer*, <u>Geneva Ecola*</u> and Zerina Kapetanovic <i>IEEE Journal of Microwaves</i>
2025	SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar <u>Geneva Ecola</u> , Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic <i>ACM SenSys</i>

Professional Experience

Summer 2024	Apple , Radio Frequency Intern Evaluated impact of chip packaging variants on radio receiver performance and automated testbed for optical chip-fiber alignment.
Summer 2023	Tesla , Wireless Hardware Intern Evaluated localization technologies for a passive entry system including BLE and UWB. Designed and evaluated RF balun circuit matching between chip and antenna.
2019 – 2021	Trellisware Technologies , Senior Radio Frequency Engineer Redesigned the high-frequency analog front-end of radios using simulations and tested hardware in lab. Led hardware sustainment of a software-defined radio digital board, coordinating team of 10+ engineers.
Summer 2018	Echodyne , Antenna Intern Measured electronically scanning array antenna near field at multiple scan angles and frequencies and developed MATLAB scripts to quantify antenna performance metrics.

Fellowships

2026	Stanford Dissertation Completion Fellowship
2024	Stanford Graduate Fellowship, Rambus Corporation Fellow
2024	Apple-Stanford Integrated Systems PhD Fellowship <i>Proposal:</i> Enabling Ultra-Low Power Smart Sensing Systems
2021	Stanford Graduate Fellowship, Rambus Corporation Fellow

Honors and Awards

2026	Best Presentation at ACM MobiSys Rising Star Forum
2026	Best Poster Runner Up at ACM MobiSys N2Women Workshop
2026	Qualcomm Innovation Fellowship Finalist <i>Proposal:</i> Ultra-low Energy IoT Networks Using Thermal Noise Communications
2025	Stanford Emerging Environmental Scholar Research Award <i>Proposal:</i> Bringing Smart Sensing Systems Everywhere on Earth
2023	Stanford EE315 Best Analog to Digital Converter Design
2019	Harvey Mudd College Departmental Honors in Engineering
2018	Harvey Mudd College T. Larry Norin Memorial Scholarship
2018	Harvey Mudd College Best Engineering Clinic Presentation
2015	Harvey S. Mudd Merit Award

Talks

- 2026 **Harnessing Ambient Radio Signals for Ultra Low Power Sensing and Communication Systems**
ACM MobiSys Rising Star Forum
- 2026 **Ultra-low Energy IoT Networks Using Thermal Noise Communications**
Apple Fellowship Talk
- 2025 **Doing More with Less: Harnessing Ambient Signals for Wireless Sensing and Communication**
University of Washington Department of Electrical Engineering
Apple Fellowship Talk
- 2025 **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
ACM SenSys
Apple Fellowship Talk

Posters

- 2026 **Ultra-low Energy IoT Networks Using Thermal Noise Communications**
Geneva Ecola, Jida Zhang and Zerina Kapetanovic
ACM MobiSys N2Women Workshop
- 2026 *Qualcomm Innovation Fellowship Finals*
- 2025 **Vision Assisted Beamsteering for Full Control of Wireless Power Transfer and Joint Communication Link**
Jasmin Falconer, Geneva Ecola, and Zerina Kapetanovic
Stanford Center for Image Systems Engineering Affiliates Annual Meeting
- 2025 **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
Geneva Ecola, Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic
ENSsys Workshop: Energy Harvesting & Energy-Neutral Sensing Systems
- 2024 *Stanford Center for Image Systems Engineering Affiliates Annual Meeting*

Teaching

Coursework

- 2026 **Course Developer, Internet of Things, Stanford University**
Developed laboratory exercises and a final project that teach students to build connected devices using a range of wireless communication protocols.
- 2025 **Course Assistant, Circuits I, Stanford University**
Led laboratory sections and office hours for more than 30 students in an introductory circuits course. Delivered midterm and final review lectures.
- 2023 **Course Developer and Assistant, 3D+ Imaging Sensors, Stanford University**
Led laboratory sessions and office hours for more than 30 students on ultrasonic sensors, time-of-flight cameras, and mmWave radar. Designed a final project in which students developed a through-wall imaging system using mmWave radar.
- 2017 – 2018 **Teaching Assistant, Systems Engineering, Harvey Mudd College**
Taught foundational concepts for laboratory experiments and guided students through system debugging and troubleshooting.

Service

Professional Service

- 2026 – Present **Chair and Founder, Stanford IEEE Microwave Theory & Technology Branch**
Founded the student branch and organize technical workshops, invited talks, and networking opportunities to build a community of students interested in microwave and wireless.
- 2026 **Technical Program Committee Member, ACM MobiCom S3 Workshop**
- 2025, 2026 **Reviewer, IEEE Journal of Microwaves**
- 2025 **Poster Program Committee Member, ACM SenSys**

University Service

- 2023 – 2025 **Vice President, Stanford Women in Electrical Engineering**
Organized professional development, community-building, and networking events for an organization with more than 100 members.
- 2024 – 2025 **Student Life and Culture Committee, Stanford Electrical Engineering**
Represented graduate student perspectives in departmental initiatives related to student experience, community, and culture.
- 2022 – 2023 **Outreach Chair, Stanford Women in Electrical Engineering**
Organized a mentorship program for more than 50 participants and led hands-on fruit-battery demonstrations at Stanford community engagement events.